1 Workload and dependencies

Compact: consecutive 64-byte lines

Scattered: one line per 4 KiB page

Illustrative addresses; the recorded data footprint is 8 / 16 / 24 KiB in either layout.
Scattered node i: base + i·4096 + (i mod 64)·64; full-ring warming precedes timing.

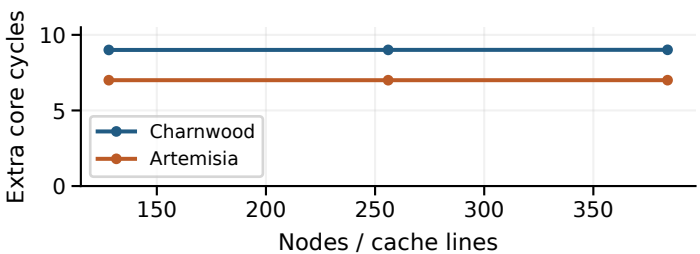
2 Controlled comparison

128 / 256 / 384 identical cache-line counts; random cycle; page-aligned 4 KiB backing.

3 Timed batch and normalization

I=100,000 dependent loads; native cost = Δ(core cycles/load) / Δ(STLB services/load).

4 Measured response



5 Inference and control

A compact baseline removes hit cost. Qualified native counters show STLB service with few cache misses/walks. The event-normalized difference estimates added L1-DTLB miss cost. Unverified protected counter coverage limits SGX to a layout timing contrast.